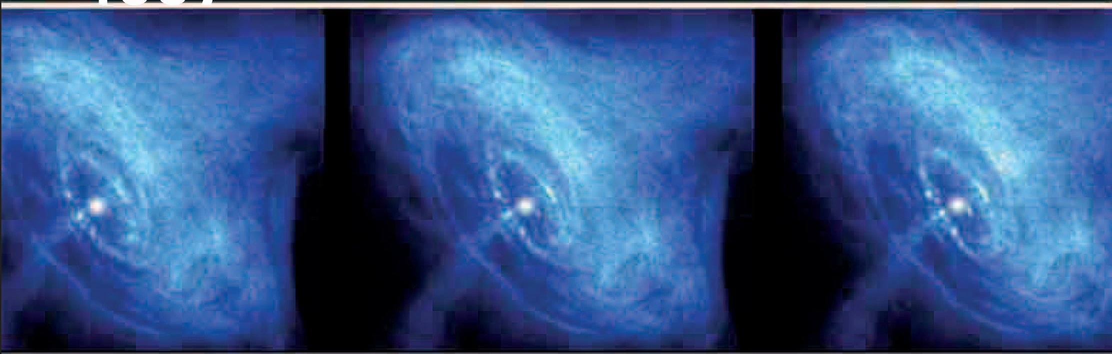
1967



First detected by their radio signals, pulsars are now known to emit other forms of radiation, such as visible light and, as shown here, X-rays.

ALTHOUGH IT WAS PRACTICALLY IGNORED at the time, American physicist **Steven Weinberg's** (b.1933) 1967 study of the forces of nature eventually became one of the most quoted scientific papers. In it, Weinberg explained how electromagnetism and weak nuclear force were just different aspects of a **single unified set of "electroweak" forces**. He also proposed that breaking the symmetry of these forces provided particles with a fundamental property—their mass. For his work, Weinberg went on to win the 1979 Nobel Prize in Physics, with Pakistani nuclear physicist Abdus Salam (1926–96) and American theoretical physicist Sheldon Glashow (b.1932).

In November, British astronomers **Antony Hewish** (b.1924) and **Jocelyn Bell Burnell** observed pulses of radio signals coming from a fixed position in the sky. They whimsically called them **LGM-1 (Little Green Men-1)**. It was later found that the signals were coming from the radiation beam of a rotating neutron star (a dense, compact star thought to be composed mainly of neutrons), and each pulse corresponded to a single rotation. In 1968, these stars were termed pulsars.

In December, surgeon **Christiaan Barnard** (1922–2001) performed **the world's first successful heart transplant** at Groote Schuur Hospital in Cape Town, South Africa. The patient, with diabetes and incurable heart disease, received a heart from a young road-accident victim. The recipient survived for just over two weeks, ultimately dying of pneumonia. Barnard was nonetheless celebrated for his achievement, and went on to perform other similar operations. His longest-surviving recipient went on to live for 23 years.

First heart transplant
South African surgeon Christiaan Barnard shows a chest X-ray image of 54-year-old Louis Washansky, the first person to undergo a successful heart transplant.

JOCELYN BELL BURNELL (b.1943)

British astrophysicist Jocelyn Bell Burnell came to prominence as a postgraduate, when she discovered radio pulsars with her thesis supervisor, Antony Hewish. Controversially, Bell did not share Hewish's 1974 Nobel Prize for this work. More recently, she served for two years as president of London's Institute of Physics.

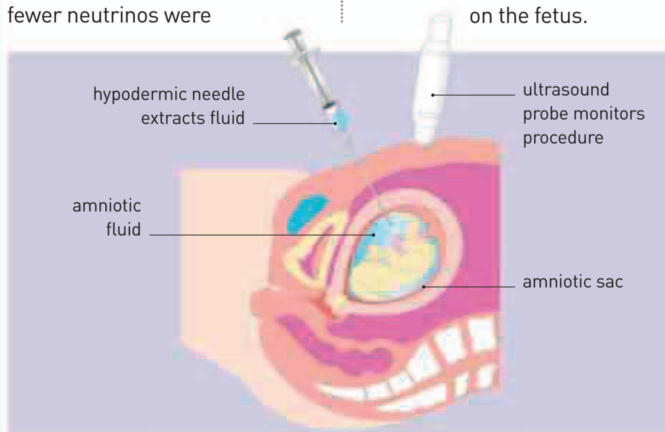
1968

This image shows tracks made by neutrinos—the most abundant subatomic particles in the universe—caught in a nanosecond inside a bubble chamber.

THE STANFORD LINEAR ACCELERATOR CENTER in California houses the longest linear accelerator—2 miles (3.2 km) long. In 1968, it provided the **first evidence for the existence of fundamental particles called quarks** (see 1963) by bombarding and shattering subatomic particles.

A solution to the **solar neutrino problem** was proposed in 1968. Neutrinos—subatomic particles with no charge and negligible mass—are generated by atomic reactions in the Sun. However, fewer neutrinos were recorded striking Earth than had been estimated. Italian physicist **Bruno Pontecorvo** (1913–1993) accounted for this discrepancy by proposing that neutrinos had appreciable mass, which allowed them to change types. Many went undetected by neutrino detectors, which monitored only one type.

American physician **Henry Nadler** (b.1936) reported on the **first prenatal diagnosis of Down syndrome using amniocentesis** (see panel, below). His observations were confirmed by direct diagnosis on the fetus.



AMNIOCENTESIS

The amniotic fluid, which surrounds a fetus, contains cells that come from the unborn child. Amniocentesis is a method by which a sample of this fluid is extracted to test for abnormality. In 1952, British obstetrician Douglas Bevis (1919–94) discovered how it could be used as a diagnostic tool. By the 1960s, scientists could use it to detect chromosome abnormalities, including Down syndrome.

British geophysicist Dan McKenzie and American geophysicist W. Jason Morgan **describe tectonic plate movements**

August 6 Irish physician Frank Pantridge describes **mobile intensive-care unit** with defibrillator; he is regarded as the "father of emergency medicine"

November 20 Steven Weinberg presents his model of **unification of electromagnetism and weak nuclear forces**

November 28 A **pulsar** is observed for the first time

December 3 Christiaan Barnard performs the **first successful heart transplant**

December 6 American surgeon Adrian Kantrowitz performs the **first child heart transplant**

December 29 American physicist John Wheeler **popularizes the use of the term "black hole"**

The Stanford Linear Accelerator Center, US, finds **evidence for the existence of quarks**

American physicist Richard Feynman describes **theory of partons**, later recognized as quarks and gluons

July 18 Semiconductor chip manufacturer **Intel founded** in California

The first **experiments to capture solar neutrinos** undertaken

September 15–22 Soviet Zond 5 becomes the first spacecraft carrying **animals to circle the Moon and return safely to Earth**

December 1 Henry Nadler uses **amniocentesis to diagnose Down syndrome** prenatally for the first time